

ERC Consolidator Grant 2024

Part B2

Section a: State-of-the-art and objectives

a.1 State of the art

Mining regions play a critical role in sustainable development. On the one hand, they help tackle global challenges by providing the mineral resources necessary to support green energy and the digital transition worldwide, but on the other hand, these regions will have to manage a wide range of **local adverse risks and impacts** of mining¹. The work of myself and others has depicted a rapid expansion of mineral extraction **to supply global markets** within regions that are critical for the conservation of biodiversity and climate stability^{2–8}. 79 % of the global extraction of metal ores in 2019 originated from five of the six most species-rich biomes, with production volumes doubling since 2000 in tropical ecosystems². The mining of metal ores and non-metallic mineral fragments degrades and removes natural habitats and is characterised by high energy and water requirements, as well as high levels of pollution and waste generation^{9–11}. In addition to increasing environmental pressures and impacts, social conflicts intensify with the expansion of raw material extraction activities, with clear evidence of the increasing number of conflicts related to mining projects^{12–14}. Furthermore, regions that extract resources must deal with the long-term legacy of mining waste dumps, which pose major risks of environmental pollution and disasters^{15–23}. Therefore, the **local impacts and risks of extractive industries** (e.g. land use changes, waste flows, and pollution) **need to be monitored globally** to enable the implementation of relevant regulatory initiatives to support sustainable development.

Assessing impacts and risks of mining requires spatially and temporally consistent data. At the same time, it requires local relevance and contextualisation because of geographically heterogeneous environments and socio-economic conditions. However, **there is no comprehensive geospatial data on mining activities that provides public access to up-to-date information at the individual production sites**^{24–26}, not even for the European Union (EU)²⁷. Geospatial information on mining is scattered, and when it exists, it is difficult to find and compile^{27–29}. Mining companies provide primarily aggregated data at the corporate level instead of reporting at the level of individual operations²⁵. For example, according to the Global Reporting Initiative (GRI), in 2017 only 216 companies voluntarily produced sustainable performance reports, but most of them only provided information and data at the company level²⁵. In a recent study³⁰, we found that even large mining companies, such as Rio Tinto Group, only provide aggregate sustainability reports. Moreover, reporting variables necessary for sustainability performance assessment (e.g. water and land use, waste generation, and land pollution) are entirely voluntary and so far not enforced by stock markets or national authorities. Open and transparent data on individual mining operations would be of high value to science and promote trust among stakeholders^{24,25,27}. However, despite the public pressure of powerful investor groups and initiatives to promote transparency, it is uncertain whether the mining industry will continue to retain its corporate prerogative or will endorse **public disclosure of information**¹⁵.

Compilations exist, such as the S&P Capital IQ Pro database (formerly S&P Global Market Intelligence SNL Metals and Mining Database)³¹, which is the basis of nearly all top publications in the field. But this sits behind a high paywall, and it is incomplete. Recent work by myself and others point out **large discrepancies between available data and the reality** on the ground^{28,32–34}. For example, I have estimated that 56 % of the mining areas visible in satellite imagery misses the corresponding production data in the S&P database (see Figure B1.1)³⁵. It is important to note that this analysis only looked into mineral production. The lack of data on waste and other indicators is significantly larger

and more challenging, even to estimate. Nevertheless, outdated and incomplete databases are the only option available and, therefore, are widely used as the default option in several global assessments of the mining sector^{2,3,7,16,36}, possibly misleading conclusions and policies.

The increasing availability of **satellite Earth observation (EO)** data offers significant potential for filling knowledge gaps in global mining. It has enabled my colleagues and me to map the visible land footprints of mines globally through the visual interpretation of satellite images^{32,37,38}. However, the rapid expansion of the mining industry challenges this method's ability to provide timely global updates. Moreover, mapping mines' land footprint is just the first step; other factors like mining waste flows and associated minerals are urgently needed to support more comprehensive assessments of mining operations worldwide^{17,39}.

Currently, we have access to methods that can effectively produce detailed mineral maps^{40–46} and highly accurate terrain change measurements^{47–49} that can provide insights on mining waste³³. However, these methods are limited to a specific set of satellite sensors with unique characteristics, often lacking global continuous coverage or having a short operational lifespan. This limitation restricts the use of such methods and sensors for global-scale monitoring of highly dynamic processes like mining. In contrast, data-driven **artificial intelligence (AI)** methods utilising satellite sensors with global, long-term coverage have shown promising results in producing variables relevant for mine monitoring, such as mapping mining infrastructure^{50–60} and estimating heights^{61,62}. Yet, scaling up AI models to larger areas requires extensive training sets, currently unavailable for mining sites.

The MINE-THE-GAP proposes an innovative research programme that will **develop the methodological foundation to enable global mining spatial-temporal information**, revealing where and when mining occurs, how much material is displaced, and what commodities and minerals are related to individual mining operations. MINE-THE-GAP will address the following overarching research question: **How can we produce timely, locally relevant, and globally reliable mine-level indicators?** Methodologically, the project presents a breakthrough by integrating high-precision data products derived from various satellite sources to substantially enrich training sets that will enable the creation of scalable AI models to map mining indicators effectively*. This approach will, for the first time, produce global-scale mine-level indicators to support risks and impact assessments of mines and provide the basis for more comprehensive global sustainability analyses of mining regions.

a.2 Objectives

MINE-THE-GAP's aim is to develop and apply methods to produce timely basic indicators at the level of individual mining sites that are locally relevant and at the same time spatially and temporally consistent to allow for global and regional impact and risk assessments.

The following five specific objectives are defined, each clearly linked to the project's work packages:

Objective 1 Construct a global-scale AI-ready database covering more than 80,000 polygons, including detailed semantic labelling of mining infrastructure types such as open pits, overburden piles, waste rock piles, and tailings dams, terrain heights within mines and minerals and commodities information (WP 1, WP 2, and WP 3).

Objective 2 Develop accurate and scalable models to automatically detect and map mining facilities such as open pits, overburden piles, waste rock piles, and tailings dams (WP 1).

Objective 3 Develop accurate and scalable models to estimate waste generation indicators based on terrain changes within mining infrastructure types such as open pits, overburden piles, waste rock piles, and tailings dams (WP 2).

*MINE-THE-GAP will use AI algorithms exclusively to produce information about mining sites from publicly available EO data sources and, therefore, does not raise any ethical issues.

Objective 4 Develop accurate and scalable models to detect commodities and minerals in individual mining facilities (WP 3).

Objective 5 Integrate and validate the developed models into a framework to produce basic spatial-temporal indicators for individual mining facilities (WP 4).

a.3 Originality and impact

The MINE-THE-GAP project will deliver a **scientific breakthrough** by designing a methodology capable of producing timely, locally relevant, and reliable global-scale mine-level indicators. Given its **interdisciplinary nature**, the project is positioned to significantly advance research across multiple disciplines. In geoinformatics and remote sensing, the project will enhance understanding of how to strategically combine various satellite data sources, addressing a key challenge in scaling AI training for global EO applications – the small size and biased training sets. The approach introduced by the project will enable the creation of AI models that are capable of continuous improvement, harnessing the potential of various highly accurate data products from current and future satellite missions. Moreover, we will produce the largest openly available AI-ready database for mining sites that will serve as a benchmark to speed the development and testing of new AI algorithms.

MINE-THE-GAP holds **disruptive potential** in sustainability research by significantly expanding the methodologies for studying global mining using EO images, independent of company reports. This project will, for the first time, enable the identification of what is being mined, the volume of displaced materials, and the location and timing of the activities. These will overcome the existing limitations of incomplete and low-quality spatial data on mining typically extracted from reports, providing a new set of essential indicators from which further sustainability assessments can expand and build on. The overall volume and characteristics of materials displaced within mining sites, which are currently under-reported, can help improve total material footprint accounts^{17,63} and assess circular economy potential existing mining waste^{28,29}. New spatially explicit data on mining can support more accurate assessments of global supply chain impacts^{64,65} and studies on mineral resource governance²⁵, as spatial data is essential to provide geographic context and spatial relationships to other environmental and social variables. Georeferenced mining infrastructures, such as tailings dams, combined with waste volume and mineral indicators, will substantially enhance the robustness of comprehensive mining pollution and disaster risk assessments^{16,21} and help identify and monitor global mining reclamation sites⁶⁶. The new spatial indicators can increase the scope of variables to assess the impact of mining operations in the Earth System, going beyond carbon³⁰. Further, the capability to provide timely information on mine expansion will impact global land use change assessments and monitoring, as well as forest and biodiversity loss studies⁵.

MINE-THE-GAP stands to make a significant impact on transparency in the global mining sector, addressing an **urgent theme** with substantial potential for impact beyond academia, including policy, industry, finance and civil society. The project outcomes will enhance the availability of crucial information that can potentially support local communities, NGOs, and other actors in making informed decisions and holding the mining industry accountable. A better understanding of the environmental and social impacts and risks of nearby mines will empower communities to advocate for their rights and interests, while NGOs and civil society organisations can use the data to ensure environmental standards and promote more sustainable mining practices. The outcomes of the project have the potential to enhance evaluations of corporate environmental impacts and guide decisions regarding the allocation of public and private investments. Altogether, this can encourage the mining industry to foster corporate transparency and wider public discussions on mining.

The project will enable **new knowledge** on global mineral extraction that is relevant to several policy and international agreements. It can support the evaluation of goals like those set out in the Paris Agreement, the UN Sustainable Development Goals, and the Kunming-Montreal Global Biodiversity Framework (GBF), by bringing new spatial data that can be linked, for example, to carbon emissions,

biodiversity loss, ecosystem restoration potential and indigenous land rights.

The project is relevant to the European Green Deal, particularly to the aim “to improve mining projects’ environmental impact in the EU area and to support similar goals in countries outside the EU”⁶⁷. EU policy initiatives, such as the Corporate Sustainability Reporting Directive (CSRD) and the Corporate Sustainability Due Diligence Directive (CSDDD), aim to provide transparency on impacts throughout global value chains and mitigate risks related to raw material sourcing. As an estimated 49,000 companies in the EU will need to report on social and environmental sustainability risks along their supply chains, I expect a rapid rise in the demand for data and analytics on mining-related impacts.

To ensure that the outcomes of the project reach a wider audience, results will be disseminated through the project’s website, high-impact journals, social media, and conferences, and all models, codes, and data will be freely available (e.g., on Github at www.github.com/mine-the-gap). We will adopt FAIR data-sharing principles to ensure Findability, Accessibility, Interoperability, and Reusability⁶⁸. For example, we will store data and codes in persistent repositories under an open license alongside comprehensive metadata, data collection protocols, as well as usage notes and adhere to robust data encoding standards. To further increase the impact of the project, I will share regular updates of the research progress and new findings with my international networks in academia and the private sector.

Section b: Methodology

b.1 Structure of the project

As illustrated in Figure B2.1, MINE-THE-GAP will be implemented in four work packages (WP), each designed to deal with one dimension of mining indicators. WP 1 focuses on mining land use indicators and will develop scalable models to map a variety of infrastructures that compose mines. WP 2 will add a third dimension to the maps produced in WP 1 by developing models to map indicators of mining waste generation. WP 3 will develop scalable models to identify materials associated with the mapped mines. Each work package will also produce a large training set with variables associated with their respective domain. Finally, WP 4 will create a framework that combines and validate the key innovations realised in WP 1, WP 2, and WP 3 to produce timely, locally relevant, and globally consistent mine-level indicators.

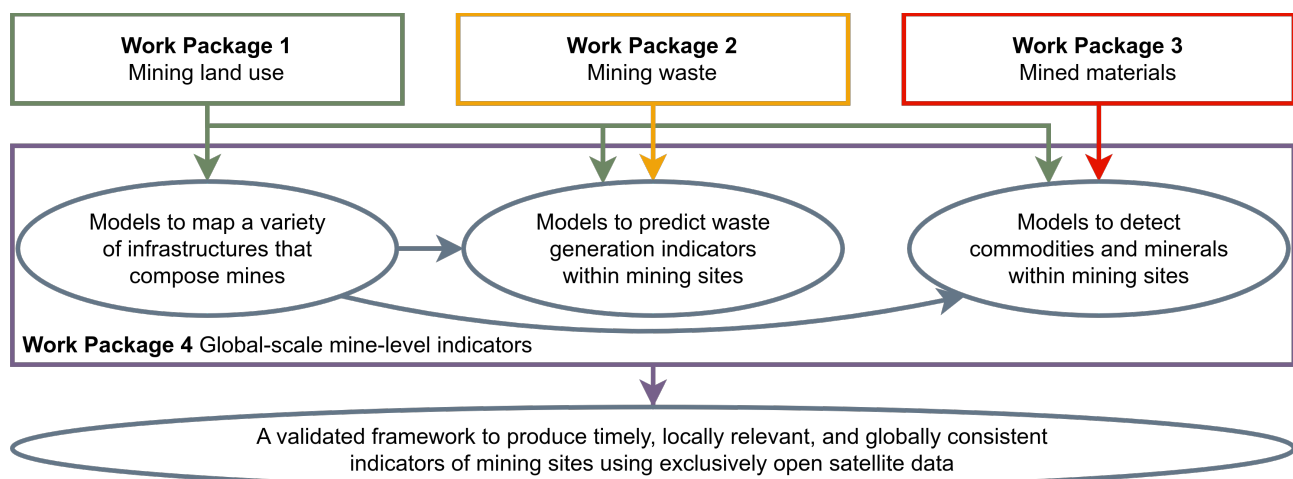


Figure B2.1: Structure of the MINE-THE-GAP project.

WP 1: Mining land use (years 1-3)

Objectives and research questions

The initial step in generating mine-level indicators involves locating mines and identifying their

terrestrial footprints. Consequently, **WP 1** is centred around two main objectives. The first is to assemble the most extensive, openly available, AI-ready database of georeferenced mines, including labelled infrastructure types. The database will cover ground features visible on the Earth's surface from satellite images, including various types of infrastructures associated with mining waste and potential sources of pollution. It will include all visible infrastructure belonging to underground and surface mining operations, which excludes underground excavations. The second objective is to develop scalable models to detect mines and map spatial features that compose the mines.

Research questions in **WP 1** include: How to design a land classification system for various mining infrastructure types that is globally applicable while also yielding locally pertinent information? What is the range of spectral responses exhibited by different mining infrastructure targets? Which ML and DL models can most effectively differentiate these mining targets within expansive, diverse regions? Additionally, is there potential for enhanced detection of various mining infrastructure types through the combined use of optical and radar imagery?

Implementation

Task 1.1: Compile a global scale AI-ready database of mining sites

To establish the AI-ready database for semantic segmentation, our starting point will be the aggregation of previous mapping initiatives led by my team^{32,69} and colleagues³⁷. These collective efforts have yielded 81,562 non-overlapping polygons. However, these polygons lack precise temporal references, as they were delineated using composite and multi-year images from various sources. Additionally, these datasets do not differentiate between types of mining infrastructure. In Task 1.1, these polygons will serve as a foundational dataset for generating an extensive set of AI-ready image patches for the semantic segmentation of mining areas, including different types of mining infrastructures, as illustrated in **Figure B2.2**.



Figure B2.2: Example of mining infrastructure types. Green = waste rock dumps, light blue = tailings storage facilities, dark blue = water storage pond, orange = milling and processing infrastructure areas, purple = open cut pits, and dark red = heap leach pads. Source: Werner et al. (2020)³⁸.

To facilitate the creation of a comprehensive global training set, we will focus on the geographical areas of the 81,562 polygons^{32,37}. We will employ an assisted-segmentation technique⁷⁰ to expedite the delineation of mining borders. The assisted segmentation will be integrated into our existing open-source tool that I have designed to delineate mining areas using a range of satellite data sources^{32,71}. This method will significantly accelerate the visual interpretation and pixel-level labelling of different mining infrastructure types, such as open pits, overburden piles, waste rock piles, and tailings dams. Drawing from my previous experience^{32,69} and statistical analyses in mine interpretation work, typically only about 10% of the effort is allocated to target identification, with the remaining 90% devoted to border delineation. Therefore, an assisted segmentation will streamline this process, effectively addressing the bottleneck in developing a large-scale AI-ready training set.

The training dataset will be organised into AI-ready image patches^{72,73}, each comprising a pair of multiband images from the Sentinel-2 Multispectral Instrument (MSI) and Sentinel-1 Synthetic Aperture Radar (SAR) sensors. Integrating these two optical and radar sensors is a strategic choice for three main reasons. First, their synergy has been proven to enhance predictions of heights^{61,62}, land cover types⁷⁴, and mining tailings⁵⁹. Second, the Sentinel missions were selected because they offer global coverage, with spatial and temporal resolutions optimal for mining operations analysis (up to 10 m and a 3-day revisit cycle, varying by spectral band and geographic location)^{32,69}. Third, Copernicus's open data policy and sustained mission planning guarantee the longevity and relevance of the **MINE-THE-GAP**'s methodologies, enabling continuous, near-real-time, global-scale monitoring of mining indicators.

Task 1.2: Develop scalable models for semantic segmentation of mining areas

Employing an assisted-segmentation approach in mapping mines is pivotal for generating ample data to train scalable segmentation models. However, this method alone is insufficient for regularly updating the global land footprint of both existing and emerging mining sites. Therefore, we will work towards developing a fully automated method for detecting and mapping mining infrastructure. The existing literature supports the capability of AI algorithms, particularly those based on convolutional neural networks (CNNs), to accurately identify open pit mines^{50,53–58}, tailings dams^{51,52,59,60}, and artisanal small-scale mining operations⁷⁵. CNNs, unlike pixel-based methods, excel in interpreting spatial context, edge placement, and relative object size⁷⁶, which are crucial for identifying mining infrastructures⁷⁷. However, expanding the area of applicability of AI models to larger areas poses challenges, mainly due to the small spatially biased training sets⁷⁸ and the difficulty in distinguishing mining sites from other land cover types using only optical image data⁵⁹.

The newly developed database in Task 1.1 will address these limitations by providing large and diverse training sets and a multi-modal predictors space. This enhancement will propel us beyond current methodologies, enabling the exploration of various network architectures and developing scalable models for mine detection and mapping. In addition, we will integrate other publicly available training sets for land cover^{72,73} and utilise Sentinel scene classification⁷⁹ to enrich the representation of non-mining land cover types in our training datasets. This strategy is designed to aid the models in differentiating mining areas from other land cover types, thereby reducing the incidence of false positives.

Outcomes

Task 1.1 will fulfil a significant portion of **Objective 1** by creating an unprecedented, multi-modal, AI-ready database for the semantic segmentation of mining areas, encompassing over 80,000 sites. This database will be made publicly accessible under an open license, complemented by a detailed data descriptor article. Task 1.2 will generate insights into the efficacy of various combinations within the predictor's space and evaluate different CNN architectures for mining area mapping. This task will also lead to the development of scalable models specifically designed for mapping key mining components such as open pits, overburden piles, waste rock piles, and tailings dams. This will mark the completion of **Objective 2** and provide essential support for the initiatives outlined in **WP 4**. Collectively, **WP 2** is expected to yield a minimum of four peer-reviewed academic articles, which may be integral to the composition of a PhD thesis.

WP 2: Mining waste (years 1-3)

Objectives and research questions

WP 2's first objective is to integrate terrain height data as an additional layer into the AI-ready database developed in Task 1.1. This integration will facilitate the central goal of **WP 2**, to develop scalable models for mapping waste generation indicators within mining sites. These models will produce

time series of terrain height reflecting changes due to raw material excavations and waste material deposition.

Key research questions in **WP 2** are: Among the various ML algorithms, which ones demonstrate superior performance in predicting variables associated with mining waste? How effectively can ML predict these variables for large-scale mapping across diverse landscapes and mine types? Is there potential to enhance the accuracy of height and depth predictions within mining areas through combined optical and radar images? Furthermore, which indicators serve as the most reliable predictors of waste volumes?

Implementation

Task 2.1: Integrate terrain information into the AI-ready database

While the predictors of AI models will be solely extracted from open satellite images with global long-term coverage, training sets will be assembled using several satellite data products. Task 2.1 is dedicated to expanding the variables in the AI-ready database by incorporating terrain elevation data from multiple sources. A key focus will be on utilising the suite of three-dimensional (3D) data produced by NASA's spaceborne Light Detection and Ranging (LIDAR) missions: ICESat-2⁸⁰ and the GEDI⁸¹. These open data products offer near-global coverage from 2018 to 2023 and represent a valuable resource for studying terrain changes. They provide precise terrain elevation data, with a mean error of less than 1 meter in areas of sparse vegetation⁸². While these missions are not planned for the long term, they are instrumental in building large training sets for developing models. These models will predict terrain heights using alternative satellite data sources with global, long-term coverage, such as the Sentinel missions.

Additionally, we aim to augment the training set with high-resolution Digital Elevation Models (DEMs) derived from Synthetic Aperture Radar Interferometry (InSAR) processing of TanDEM-X and TerraSAR-X images. These DEMs are accurate^{83,84}, but their application is limited to select locations due to the limited availability of free images, which are provided for research purposes upon request⁸⁵. Although TanDEM-X and TerraSAR-X satellites do not offer continuous global time series, they have enabled an open TanDEM-X 30m Edited DEM data^{86–88}. This dataset will be particularly useful in areas where image acquisition dates coincide with those of Sentinel-1 and Sentinel-2 missions. Furthermore, open DEMs from specific countries, like the USGS's 3D Elevation Program⁸⁹, will serve as valuable resources for both training and validating our models. This strategic approach ensures the generation of a substantial and diverse training dataset on terrain height within global mining areas.

Task 2.2: Develop scalable models to estimate indicators of waste generation

The terrain data products mentioned earlier are capable of providing accurate height measurements. When two such datasets, acquired at different times, are available for an area, they enable deriving precise height changes maps^{47,48}, as depicted in **Figure B2.3**. Comparing DEMs from different dates in mining areas is a promising method for estimating the volume of mining waste³³. However, the current production frequency of DEMs does not align with the dynamic nature of the mining sector, limiting their utility for ongoing waste generation monitoring. To address this, Task 2.2 will focus on developing ML models that can efficiently generate information on the heights of various mining infrastructure types as outlined in **WP 1**.

A predictor's space, combining radar backscatter with optical spectral reflectance and encompassing multi-temporal, multi-spectral, multi-polarised, and spatial context data, has already proven effective in producing accurate height estimates for urban areas on a large scale^{61,62}. The extensive training set on terrain heights, which we will compile in Task 2.1, will enable training scalable models to track terrain alterations within mining sites. These terrain changes, in turn, will serve as a foundational metric for estimating the volume of overburden materials, tailings, and overall volumetric alterations.

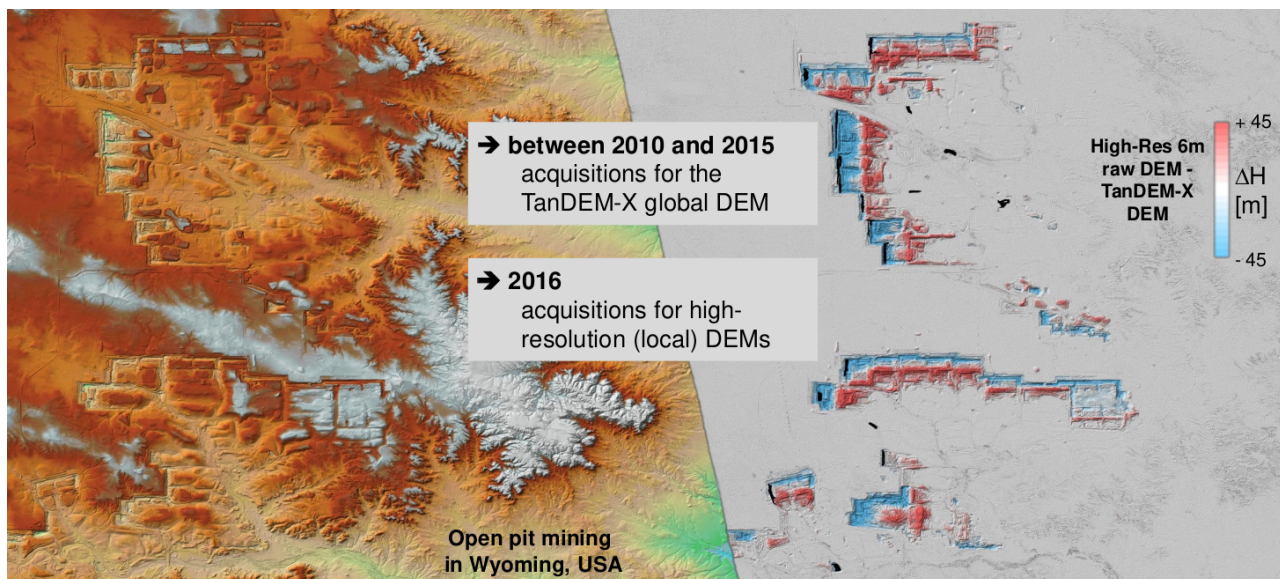


Figure B2.3: Illustration of a high-resolution digital elevation model (left) and terrain elevation changes (right) within an open pit mine in Wyoming USA. Source: Dr Marie Lachaise, German Aerospace Centre (DLR).

Outcomes

Task 2.1 will enhance the AI-ready database, contributing to achieving **Objective 3**. Its outcomes will be openly shared, including the augmented database and a data descriptor article. Task 2.2 will fulfil **Objective 3** by developing ML models to estimate mining waste indicators based on height changes in various mining infrastructures. These tasks are expected to yield outcomes significant for at least four peer-reviewed articles, contributing to assembling a PhD thesis.

WP 3: Mined materials (years 3-5)

Objectives and research questions

WP 3 is set to achieve two dynamic objectives. Initially, it will enrich the AI-ready database with detailed information on commodities and minerals. This enhancement is pivotal as it sets the stage for the second goal: develop ML models adept at identifying the specific commodities and minerals linked to the mining infrastructures identified earlier.

Research questions in WP 3 include: Is there a global consistency in the spectral response patterns of different mining types? Do mining features exhibit uniform spatial patterns worldwide? Which machine learning models demonstrate the highest accuracy in detecting specific commodities and minerals associated with mining infrastructures? How can we develop and refine a commodity group classification system that is compatible with the parameters measurable through Earth Observation (EO) data, enabling CNN algorithms to classify satellite images into specific commodity groups? Furthermore, we seek to understand if synergies between optical and radar imagery can significantly improve the detection and identification of commodities and minerals.

Implementation

Task 3.1: Integrate minerals information into the AI-ready database

This task will develop a new training set for mineral presence detection, utilising automated hyperspectral mineral identification techniques⁹⁰. This approach effectively identifies minerals across extensive regions where field access is limited⁴⁰. We will employ hyperspectral images from the EnMAP⁴¹ and Prisma⁹¹ missions to map minerals within selected spatial features from Task 1.1. However, these satellites do not offer continuous global coverage. The images from EnMAP and

Prisma are freely available in a limited number upon request^{91,92}. We will use these images, alongside automated hyperspectral mineral identification⁹⁰ and open spectral libraries (Table B2.1), to create datasets within mining areas. Additionally, data from NASA's EO-1 - Hyperion⁹³ will allow covering several locations around the world as we find a large overlap between available EO-1 data and mining sites identified in previous work^{32,37,69}. Information on commodities and commodity groups, extracted from company reports⁹⁴ and proprietary datasets³¹, will also be added to the AI-ready database. This structured approach ensures the production of an extensive training set, which will be essential for developing robust models.

Table B2.1: Open spectral libraries.

Library	Number of spectra
GFZ spectral reference library ⁹⁵	86
ECOSTRESS Spectral Library ⁹⁶	3,400
USGS Spectral Library ⁹⁷	2,468
SPECCHIO Spectral Library ⁹⁸	4,802
LUCAS SOIL Spectral Library ⁹⁹	>40,000
Open Soil Spectral Library (OSSL) ¹⁰⁰	>110,000 (Compilation of several sources)

Task 3.2: Develop scalable models to detect minerals

This task prioritises mineral identification, where hyperspectral data are essential for accurately differentiating surface mineralogy¹⁰¹. While certain minerals, like iron minerals, can be identified using multi-spectral data through absorption feature-based indices¹⁰¹, the integration of multiple bands, sensors, and seasonal variations can further enhance mineral detection^{102–105}. Consequently, we will investigate various combinations of predictor space, including multi-spectral and radar data, which may reveal additional possibilities for identifying minerals beyond hyperspectral data.

Task 3.3: Develop scalable models to commodity groups

Task 3.3 involves employing CNN for commodity detection. This strategy differs from pixel-level mineral mapping as the CNN model will classify larger image patches based on a commodity (or commodity group) classification system. Identifying individual commodities using multi-spectral images can be challenging due to their co-occurrence in deposits and extraction as by-products¹⁰⁶. However, from my experience with satellite image interpretation, commodity groups are often discernible through spatial patterns, reflectance, and context. Therefore, establishing a refined commodity classification system will enable CNN algorithms to categorise images into distinct commodity groups accurately.

Outcomes

Task 3.1 will generate precise training sets focusing on minerals and commodities, completing the **Objective 1**. These sets will lay the foundation for creating scalable models in Tasks 3.2 and 3.3, thereby achieving **Objective 4**. The models developed will play a crucial role in **WP 4** in characterising displaced materials within mining sites. This work package is poised to make significant academic contributions, anticipated to result in the publication of five peer-reviewed articles. This includes a detailed data descriptor for the newly developed training set. In line with the other work packages, these scientific outcomes will also be part of a PhD thesis.

WP 4: Global-scale mine-level indicators (years 3.5-5)

Objectives and research questions

The objective of **WP 4** is to integrate and validate the models developed in the project into a single

framework. This will facilitate reusability and access to methods and data for further research in sustainability assessments of mining. Research questions in WP 4 include: What is the accuracy of the models on a global scale? What is the area of applicability of the models? What is the spatial distribution of the prediction uncertainty?

Implementation

Task 4.1: Framework design for mine-level indicators

Task 4.1 will integrate the findings from previous work packages to establish a unified framework for analysing global mining regions. Central to this task is the application of models from WP 1 for identifying mining infrastructures, which sets the spatial context for the subsequent application of models from WP 2 and WP 3. These models will enrich the framework with data on waste generation, minerals, and commodities. The process and interconnections of these applications are depicted in the directional flows within WP 4 as shown in Figure B2.1. This integrative approach will facilitate generating a detailed time series analysis of mining sites, capturing data from circa 2015 to the latest available Sentinel 1 and 2 image pairs. The framework will be designed to provide annual updates on mining indicators, subject to regional variations and the availability of clear imagery.

Furthermore, we anticipate that future satellite missions will yield new and more accurate radar, hyperspectral, and LIDAR data products. Additionally, the field of AI is rapidly evolving, with new algorithms emerging regularly. Therefore, we want to ensure that the framework remains adaptable and scalable, enabling the easy integration of new models developed using cutting-edge algorithms or trained with novel data sources. This approach is crucial for the continuous improvement and relevance of the framework, keeping it at the forefront of advancements in the field.

Task 4.2: Validate mine-level indicators against independent data sets

In Task 4.2, we will extend beyond standard AI model development practices, such as generating performance metrics through cross-validation¹⁰⁷, to include validation of our models against independent data sources. The validation data will be compiled from several data open and proprietary data sources. For mining land use, Google Maps, Open Street Maps¹⁰⁸, and high-resolution images from NICFI¹⁰⁹ will be essential validation sources, as they allow to identify the different types of mining infrastructures.

Several existing data compilations will enable the validation of waste and materials. The primary data source includes our global database encompassing 1,171 mines and detailing the extraction of 80 different materials from 2000 to 2021, compiled from over 2,000 mining company reports^{94,110} (see Figure B2.4). This data collection will be extended with data from the Mineral Resources Online Spatial Data (>100,000 entries)¹¹¹, the S&P database³¹ (>34,000 properties), the Global Coal Mine Tracker (4,300 entries)¹¹², and other spatial data on mining level operations available from research articles^{38,59,113–115}. Information on the commodities and minerals are abundant on this data sources. Mining waste, on the other hand, is not so often available. Therefore, the validation waste indicators will be viable using other relevant variables, such as ore production tonnages, mill-head grades, stripping ratios, and concentrator recovery rates³⁹. Mining land use will also be used to check the overall consistency of mining waste estimates and point to possible discrepancies as these variables are strongly correlated³⁸.

Additional checks on tailings will be possible from some locations worldwide. For that, we will use additional data from the Tailings Database and Portal¹¹⁶, which provides 1,863 tailings dam coordinates, including height, volume, construction year, and construction method. Another set of 1743 tailings facilities¹¹⁷ (some of which may overlap with the Tailings Database and Portal) will also contribute to validating tailings location and dimensions. The new TanDEM-X 30m DEM Change data product^{47,48} will be a key source to validate terrain changes over time.

Furthermore, national and subnational datasets^{118–121} will enable checks to identify countries or regions that require further revision due to discrepancies. In addition, a broad network of collaborators will contribute to the validation process, offering insights based on their experience and sharing additional data collections, including ground data (detailed in subsection b.4).

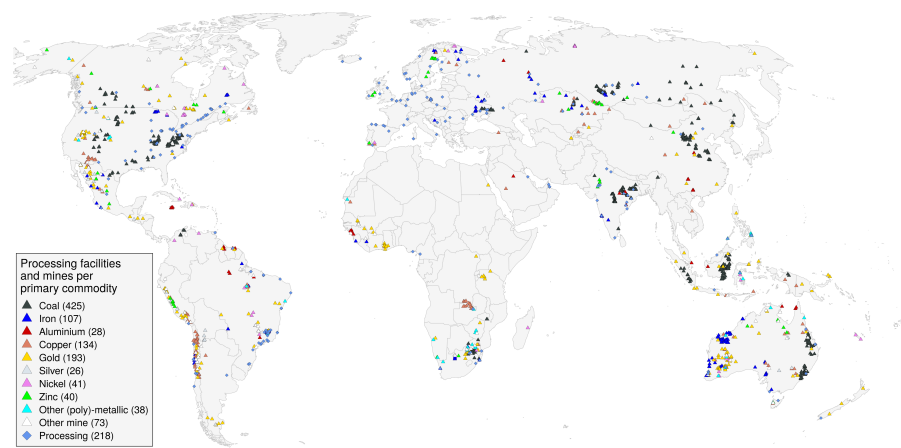


Figure B2.4: Global map of facilities covered in the open database on global coal and metal mining⁹⁴. Mines are categorised by their primary commodity. A Robinson map projection is used.

Outcomes

Task 4.1’s primary outcome is a globally consistent and validated framework designed to generate spatial data for assessing the risks and impacts of mining sites, encompassing indicators of land use, waste generation, commodities, and minerals. This achievement will successfully accomplish the final objective (Objective 5) of MINE-THE-GAP. Furthermore, at least two peer-reviewed publications are expected from the outcomes of this work package addressing the above-mentioned researcher questions.

b.2 Time table

MINE-THE-GAP will be carried out in four work packages and will run for five years. The following Table B2.2 illustrates the specific tasks within each work package and the related running times. Eight milestones are also specified, which illustrate key intermediate measurable achievements.

Table B2.2: Gantt chart of the MINE-THE-GAP project.

Work Packages/Tasks	Years									
	Months		1	2	3	4	5			
	1-6	7-12	13-18	19-24	25-30	31-36	37-42	43-48	49-54	55-60
WP 1: Mining land use										
Task 1.1: Compile a global scale AI-ready database of mining sites										
Task 1.2: Develop scalable models for semantic segmentation of mining areas						M2				
WP 2: Mining waste										
Task 2.1: Integrate terrain information into the AI-ready database		M1								
Task 2.2: Develop scalable models to estimate indicators of waste generation					M3					
WP 3: Mined materials										
Task 3.1: Integrate minerals information into the AI-ready database						M4				
Task 3.2: Develop scalable models to detect minerals							M6			
Task 3.3: Develop scalable models to detect commodity groups									M7	
WP 4: Global-scale mine-level indicators										
Task 4.1: Integrate models for land use, waste, and materials										
Task 4.3: Validate mine-level indicators against independent data sets							M5			M8

- M1 (Month 12): First version of the AI-ready database including semantic segmentation and terrain data
- M2 (Month 36): Models to map various mining infrastructure
- M3 (Month 36): Models to predict indicators of mining waste
- M4 (Month 36): Second version of the AI-ready database including minerals and commodities
- M5 (Month 42): Validated framework for area and waste generation indicators
- M6 (Month 48): Models to detect minerals
- M7 (Month 60): Models to detect commodity groups
- M8 (Month 60): Validated framework for materials and commodities indicators

b.3 Host institution

The MINE-THE-GAP project will be based at the Vienna University of Economics and Business (WU), my current home institution. WU is a public university committed to excellence in research and research-led teaching. WU's Institute for Ecological Economics, within the Department of Socioeconomics, is a leading institution for research and teaching on sustainability. The research area of geoinformatics for sustainable development, which I lead, develops cutting-edge research on machine learning and applications for monitoring global mining and related impacts. Our team has established itself internationally as a prominent group in the research area of sustainable extraction of natural resources, with a growing network of academics, NGOs, and private sector collaborations worldwide. The interdisciplinary approach of the Institute and the world-leading research teams of WU provide a unique intellectual environment in which the MINE-THE-GAP project can flourish successfully. The WU will also provide state-of-the-art facilities with modern hardware equipment, high-performance computing clusters, access to proprietary databases (e.g. Bloomberg, S&P database), and multimedia libraries.

b.4 Team requirement and team composition

The MINE-THE-GAP project will be carried out by a core team of six researchers: the PI (Victor Maus), two postdocs (N.N.), and three PhD students (N.N.). As the Principal Investigator of MINE-THE-GAP, I will lead and coordinate the overall research, execute the planned activities, and manage the rest of the team, including supervising the PhD students supported by the postdocs. The project team will bring together the PI's expertise in geoinformatics, machine learning, big-EO data analytics, and environmental assessment of mining areas, complemented by the expertise of the other team members, which will be essential for the project's success. In addition to steering the project as a whole, the PI will lead WP 1 and WP 4 and will have the support of a PhD student to implement the respective tasks, particularly in WP 1. Two postdocs will be hired to lead WP 2 and WP 3, with the opportunity to develop their research skills and careers. Postdoc 1 (N.N.) will be an expert researcher in radar satellite image time series analysis and will be responsible for Tasks 2.1 and 2.2 with the support of a PhD student. Postdoc 2 (N.N.) will lead WP 3 and bring expertise in imaging spectroscopy applied to minerals mapping. She/he will be responsible for implementing Tasks 3.1 and 3.2 with the support of a PhD student, who will be the main responsible for Task 3.3. The three PhD students (N.N.) joining the team will have a strong background in machine learning, earth observation, and excellent programming skills. Each PhD student will have the opportunity to develop their thesis focusing on one of the thematic work packages, WP 1 semantic segmentation of mining areas, WP 2 terrain modelling to estimate mining waste, and WP 3 minerals and commodities detection. All team members will also contribute to implementing tasks in WP 4 by integrating and validating the models developed in each work package.

External experts from teams led by Prof. Marius Apple at Bochum University of Applied Sciences, Germany, and Prof. Sidnei Luís Bohn Gass at the Federal University of Pampa, Brazil, will also contribute through subcontracts. They will focus on pre-processing large satellite data sets and gathering training and validation data. This collaboration builds on a long-standing partnership with these experts, yielding high-level research outcomes^{32,69,122}.

A wide network of international scholars will also foster an intellectual environment valuable for the project's implementation. Experts in different domains will collaborate on methodological developments, data sharing, and validation of mining indicators. Prof. Alex Lechner from Monash University Indonesia will engage in research exchanges, focusing on developing AI-ready datasets and semantic segmentation models, specifically in WP 1. He is an expert in global mining assessments and co-leads the project "Mining spatial data intelligence for ecological and social risk assessment (<https://www.monash.edu/indonesia/our-research/mining-ai-and-gis>)". Dr Marie Lachaise is an expert in InSAR processing and terrain change modelling at the DLR. She leads the development of the TanDEM-X 30m DEM Change data product^{47,48} and will be a key collaborator in WP 2, particularly in developing high-accuracy terrain information of mining areas and validating the models globally.

The team led by Prof. Martin Herold, head of the Remote Sensing and Geoinformatics section at GFZ German Research Centre for Geosciences, will collaborate in WP 3, particularly in the integration of EnMAP hyperspectral and ground data into MINE-THE-GAP approach and model's validation.

Furthermore, experts on the impacts and risks of mining with large experience in mining data collection have confirmed their collaboration in designing and validating global-scale mine-level indicators. These experts include the researchers on mining and sustainability, Dr Tim Werner at the University of Melbourne (Australia) and Prof. Michael Tost at the University of Leoben (Austria), Prof. Kai Qin, an expert in coal mining emissions at the China University of Mining and Technology (Xuzhou, China), Prof. Stefan Giljum, expert in sustainable resource management at the WU, and Dr Ian McCallum, at the International Institute for Applied Systems Analysis (IIASA), who brings several years of experience in producing and validating global land cover maps. These researchers will be specifically involved in WP 4, providing insights in assessing and validating the project's outcomes.

b.5 Feasibility

MINE-THE-GAP is an ambitious project that addresses a major knowledge gap in global mining in an original and interdisciplinary manner by integrating various methods and data sources that have not been linked before and, therefore, can pose several risks. However, pre-existing conditions and complementary measures will minimise potential threats to the project implementation and ensure the achievement of high gains with limited risks.

The successful realisation of MINE-THE-GAP requires a solid scientific foundation in the fields of geoinformatics and EO data analytics combined with knowledge and experience in the evaluation of mining activities. The profiles of the MINE-THE-GAP team members have been defined to complement my research expertise and ensure a successful implementation of the planned research. The planned team structure has at least two researchers, one postdoc and one predoc, who will be engaged in each work package to ensure high-quality outcomes. More specifically, I will recruit a postdoctoral researcher expert in radar satellite imagery for 3D modelling to lead WP 2; a postdoctoral researcher with expertise in spaceborne imaging spectroscopy applied to mineral mapping to lead WP 3 and three PhD students with a strong background in machine learning and deep learning applied to Earth observation data, each focusing their PhD research in developing a set of thematic indicators of mining. Hiring the project's personnel can also be considered a risk for the implementation. However, the PI's wide international network of scholars from key universities and research centres worldwide will ensure that job advertisements reach extremely qualified and motivated researchers to join the team.

Several building blocks that are critical to the research planned in MINE-THE-GAP are already in place, with promising results. These refer mainly to recently compiled large-scale mining datasets^{32,69,94}, which I have led and contributed, and the tool that I designed and built to collect information on mining areas⁷¹. My experience in elaborating and executing data collection plans will ensure a successful increase in the availability of AI-ready data, which is a natural next step in my research.

The access and processing of a large volume of satellite images are critical for the implementation of MINE-THE-GAP. To address the risk of not having access to enough data, I have planned to use mostly Sentinel 1 and Sentinel 2 images, as they have unlimited usage quota and are available directly from several cloud processing facilities, e.g. Google Earth Engine, Amazon Web Services, and Planetary Computer. In addition, they provide global continuous coverage and have long-term operation plans to provide open data.

To mitigate the risk of limited access to crucial satellite data sources for assembling training sets, MINE-THE-GAP predominantly relies on open data sources with extensive global coverage. This includes three-dimensional (3D) data from NASA's LIDAR missions, ICESat-2⁸⁰ and GEDI⁸¹. While images from satellites such as TerraSAR-X and TanDEM-X from DLR⁸⁵ will be used in smaller quantities due to the need for specific requests, they will complement and validate certain regions. Additionally,

we plan to use data from other national sources like the USGS's 3D Elevation Program⁸⁹ as needed. For assembling the training set on minerals and commodities, we will draw from extensive existing data collections^{38,59,113–115}. Hyperspectral satellites will be used to enhance training sets with mineral and mineral group information^{40–46}, particularly utilising EnMAP⁹², Prisma⁹¹, and EO-1 - Hyperion⁹³. The MINE-THE-GAP team, focusing on radar and hyperspectral data integration and validation, will collaborate closely with the German Aerospace Centre (DLR) and the GFZ German Research Centre for Geosciences.

The confirmed participation of researchers from globally recognised institutions significantly reinforces the feasibility and importance of this project. In WP 1, we will exchange experience and collaborate on image segmentation and mine mapping with Prof. Marius Apple from Bochum University of Applied Sciences (Germany), Prof. Alex Lechner at Monash University (Indonesia), and Prof. Sidnei Luís Bohn Gass at the Federal University of Pampa (Brazil). Dr. Marie Lachaise from the German Aerospace Centre (Germany) will contribute her expertise on terrain modelling to WP 2. For WP 3, Prof. Martin Herold at the GFZ German Research Centre for Geosciences has confirmed his collaboration, particularly on hyperspectral data analysis. Furthermore, the development and validation of global-scale mine-level indicators in WP 4 is of interest to several key researchers on mining and sustainability worldwide, including Dr Tim Werner from the University of Melbourne (Australia), Prof. Stefan Gljum at the Vienna University of Economics and Business (Austria), Dr Ian McCallum from the International Institute for Applied Systems Analysis (Austria), Prof. Michael Tost at the University of Leoben (Austria), and Prof. Kai Qin from the China University of Mining and Technology (Xuzhou, China). These confirmed collaborations will enrich the project's intellectual environment and facilitate exchanges, thereby enhancing the project's implementation and reducing risks.

Conclusion

I conceived and designed MINE-THE-GAP to be groundbreaking, pioneering research to globally map and survey variables that are critical to assess the impacts and risks of mines, independently of company reports. Currently, monitoring global mining activities is impossible due to a lack of a methodology to scale automated mapping approaches reliably. Therefore, I propose research that will investigate how to produce timely, locally relevant, and globally reliable mine-level indicators. To achieve that, I will integrate a wealth of open satellite Earth observation imagery and other geospatial data sources on mining in a novel way that substantially increases the availability of training data, enabling the development of scalable artificial intelligence to map basic mining indicators, which holds the potential for generalisation, setting the stage for scalable models in various research fields. This research deals with a complex range of datasets and algorithms, requiring a solid scientific foundation in the fields of geoinformatics and remote sensing and experience in the assessments of mining activities, which will be fulfilled by my own research profile combined with the other team members. As a result, the project will deliver a validated framework to disclose basic geospatial indicators related to mining operations, including land use, waste generation, and mined materials. By producing indicators that are relevant both at the local and global levels, MINE-THE-GAP will enable, for the first time, to explore, monitor, and eventually manage mineral resources and social and environmental risks in global mining regions.

Appendix: All current grants and on-going / submitted grant applications of the PI (Funding ID)

Current research grants:

<i>Project Title</i>	<i>Funding source</i>	<i>Amount (Euros)</i>	<i>Period</i>	<i>Role of the PI</i>	<i>Relation to current ERC proposal</i>
Global mining land use update	S&P Global Market Intelligence	162,516.00 €	2024-2024	Principal Investigator	This project focuses on updating global mining land use information, building upon my previous work and methodology ³² . The project has no scientific overlap with MINE-THE-GAP but can provide additional data to support MINE-THE-GAP's implementation. I will dedicate 60 % of my time to this project.
Global deforestation impacts of Europe's metal and coal consumption	Vienna University of Economics and Business (WU)	23.937,00 €	Jan/2024 - Dec/2024	Principal Investigator	This project has no overlap with the MINE-THE-GAP researcher questions. Research questions investigating the link between the minerals demand and deforestation could strongly benefit from MINE-THE-GAP's outcomes .
Seed funding	Vienna University of Economics and Business (WU)	10,000.00 €	Sep/2023 - Dec/2023	Applicant	This seed funding is to support the preparation of the MINE-THE-GAP proposal for submission to the ERC Consolidator Grant 2024 call.

On-going / submitted grant applications

<i>Project Title</i>	<i>Funding source</i>	<i>Amount (Euros)</i>	<i>Period</i>	<i>Role of the PI</i>	<i>Relation to current ERC proposal</i>
MINEXPLORER	ERC Proof of Concept Lump Sum Grant 2023 (ERC-2023-PoC)	150,000.00 €	2024-2025	Participant	MINEXPLORER will develop an open data platform for environmental impact analysis in mineral extraction and disseminate scientific findings. MINE-THE-GAP, on the other hand, focuses on creating new methods for mining indicator mapping. If both are funded, their synergy could amplify each project's impact through MINEXPLORER's extensive networks. If MINEXPLORER is successful, it will require 20 % of my time commitment.
4D-Mine	Research Fund for International Scientists by the National Natural Science Foundation of China	100,000.00 € (800,000 RMB)	2025-2026	Principal Investigator	The fund covers direct costs other than personal costs. My full-time position as the Principal Investigator and two PhD students will be co-funded by the host institution, the China University of Mining and Technology in Xuzhou, China. This project focuses on developing approaches to create accurate time-series data on mining waste using interferometric synthetic aperture radar. Unlike MINE-THE-GAP, 4D-Mine does not pursue scalability through data-driven methods. If funded, 4D-Mine could provide valuable insights for MINE-THE-GAP's Task 2.1. However, due to logistical constraints, if both projects are funded, only MINE-THE-GAP will proceed. If 4D-Mine is successful, it will require 60 % of my time commitment.

<i>Project Title</i>	<i>Funding source</i>	<i>Amount (Euros)</i>	<i>Period</i>	<i>Role of the PI</i>	<i>Relation to current ERC proposal</i>
RAWCLIC	HORIZON-CL5-2024-D1-01-06	Total grant: 5,000,000.00 € WU budget: 350,000.00 €	2025-2027	Participant	RAWCLIC and MINE-THE-GAP exhibit strong synergistic potential without scientific overlap. RAWCLIC could significantly benefit from methods and data from MINE-THE-GAP for assessing how climate change foresight affects the supply of primary and secondary raw materials. In RAWCLIC, I will dedicate 10 % of my time to conducting spatial assessments of mining impacts.

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